Switching

STP :Spanning tree protocol

VLAN: Virtual lan

VTP: Virtual trunking Protocol

INTER-VLAN

Switching basics

Switch act as a multiport bridge and its basic duty is to break collision domain.

Layer 2 switches and bridges are faster than routers because they don't take up time looking at the Network layer header information.

Switches look at frame's hardware addresses before deciding to either forward the frame or drop it.

Switching basics

Switches create private dedicated collision domain.

They provide independent bandwidth on each port.

Layer 2 switching provide the following:

- Hardware based
- Wire Speed
- Low latency
- Low cost.

Switching basics

Switches do not do any modification to the data packet.

They only read the frame encapsulating the packet.

This makes the switching process considerably faster and less error-prone than routing process.

Switch Vs. Bridge

Bridges are software based, while switches are hardware based because switches use ASIC (Application Specific Integrated Circuit) chips to help make filtering decisions.

A switch is basically a multiport bridge.

Bridges can only have one spanning tree instance per bridge, while switches can have many.

Switches have more number of ports.

Switch Vs. Bridge

Both poses multiple COLLISION DOMAIN but one BROADCAST DOMAIN.

Both learn MAC addresses by examining the source address of each frame received.

Both make forwarding decisions based on layer 2 addresses.

Switch functions

Address Learning:

Layer 2 switches remember the source hardware address of each frame received on an interface .

Switches enter this information into a MAC database called a forward/filter table.

Forward/Filter Decision:

When a frame is received on an interface, the switch looks at the destination hardware address and fields the exit interface in the MAC database.

The frame is only forwarded out the specified destination port.

Switch functions

Loop Avoidance:

If multiple connections between switches are created for redundancy purpose, network loops can occur.

Spanning Tree Protocol (STP) is used to stop network loops while still permitting redundancy.

Switch Address learning

When switch is first powered on, the MAC forward/filter table is empty.

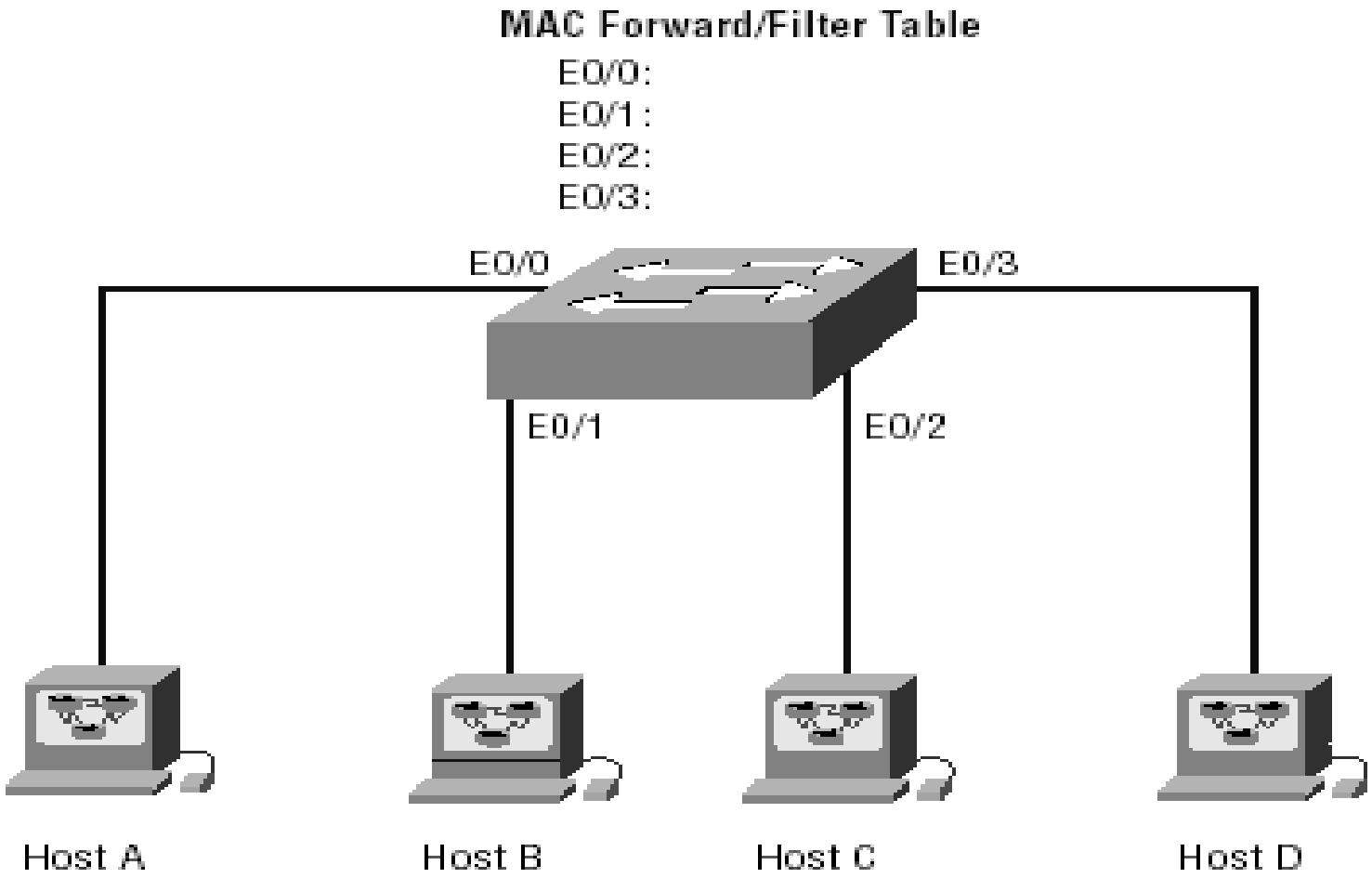
When an interface receives a frame, the switch places the frame's source address in MAC forward/filter table, allowing it to remember which interface the sending device is located on.

Switch then floods the network with this frame out of every port except the source port because it has no idea where the destination device actually located. If a device answers this flooded frame and sends a frame back, then:

- Switch takes the source address from that frame and place the mac address in the database as well.
- Switch associates this address with the interface that received the frame.

Switching table

Ξ 7.5 Empty forward/filter table on a switch

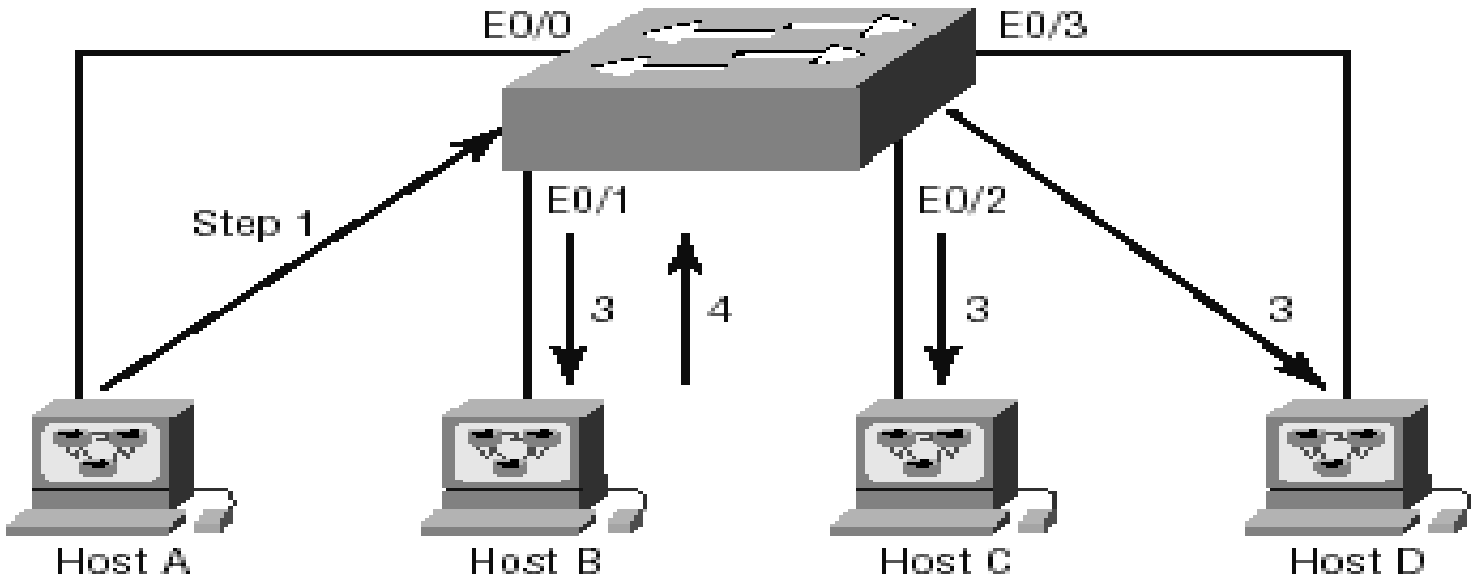


Switch addressed table

7.6 How switches learn hosts' locations

MAC Forward/Filter Table

E0/0: 0000.8c01.000A step 2
E0/1: 0000.8c01.000B step 4
E0/2:
E0/3:



STP Spanning tree protocol

Its main task is to stop routing loops from occurring on layer 2. (Bridges or Switches)

It monitors the network to find all links making sure that no loops occur by shutting down the redundant link.

It uses Spanning Tree Algorithm (STA), to first create a topology database, then search out and destroy redundant links.

With STP running, frames are only forwarded on the STP, picked links.

Switching techniques

LAN Switch Types decide how a frame is handled when it's received on a switch port.

Latency: The time switch takes for a frame to be sent out an exit port once the switch receives the frame.

There are three switching modes:

- Cut – through (Fast Forward)
- Fragment Free (Modified cut-through)
- Store-and-forward

Switching techniques

Cut-through (Fast Forward):

In this mode, the switch only waits for the destination hardware address to be received before it looks up the destination address in the MAC filter table.

Fragment Free (Modified cut-through):

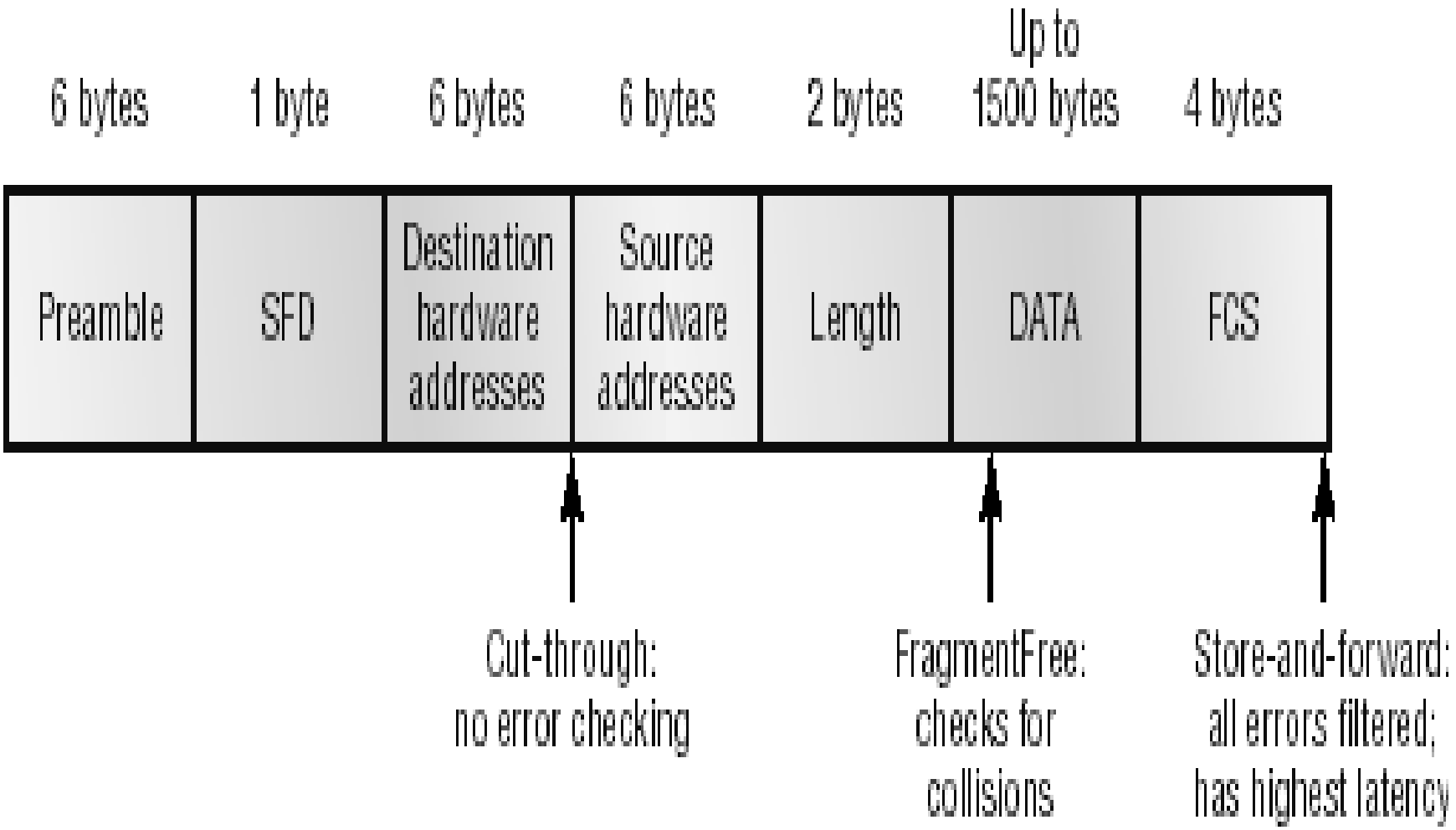
In this mode, the switch checks the first 64 bytes of a frame before forwarding it for fragmentation.

This is the default mode for catalyst 1900 series switch.

Store-and-forward:

In this mode, the complete frame is received on the switch's buffer, a CRC is run and then the switch looks up the destination address in the MAC forward/filter table.

Different switching modes within a frame



Configuring switch operations

We will cover following list of tasks:

- Setting the password
- Setting the hostname
- Configuring the ip address and subnet masks
- Setting a description on the interface
- Erasing the switch configuration
- Configuring VLANs
- Adding VLAN membership to switch port.
- Creating VTP domain.
- Configuring trunking.

Configuring switch operations

Setting user password:

switch(config)# enable password cisco

Setting privileged level password

switch(config)# enable password cisco

Console:

switch(config)# line console 0

switch(config-line)# password *cisco*

switch(config-line)# login

Telnet:

switch(config)# line vty 0 15

switch(config-line)# password *cisco*

switch(config-line)# login

switch(config)# hostname *LAN1*

Configuring Interface Description

2950 Switch:

Description command is used from interface configuration mode.

Spaces can be used within description.

```
Switch(config)# int fastEthernet 0/1
    Switch(config-if)# description Sales Printer
    Switch(config)# int f0/12
    Switch(config-if)# description trunk_to_building_4
```

VLAN`s

A VLAN is a logical grouping of network users and resources connected to administratively defined ports on a switch.

VLANs allow us to break broadcast domain in a pure switched internetwork.

VLANs allow us to create smaller broadcast domains within a layer 2 switched based internetwork.

How Vlans help to manage networks

Network adds, moves and changes are achieved by configuring a port into the appropriate VLAN.

A group of users needing high security can be put into a VLAN so that no users outside of the VLAN can communicate with them.

VLANs are independent from their physical or logical locations.

VLANs can enhance network security.

VLANs increase no. of broadcast domains and decrease the size of each broadcast domain.

Broadcast control

All devices in a VLAN are member of same broadcast domain and receive all broadcasts.

The broadcasts, by default, are filtered from all ports on a switch that are not member of the same VLAN.

This is one of the prime benefit that we get with a VLAN based switched network, otherwise we would have faced serious problem if all our users were in same broadcast domain.

Vlan security

In a flat network anyone connecting to the physical network could access the network resources located that physical LAN.

In order to observe any/all traffic happening in that network one has to simply plug a network analyzer into the hub.

Users can join any workgroup by just plugging their workstations into the existing hub.

By building VLANs and creating multiple broadcast groups, administrators can now have control over each port and user.

Since VLANs can be created in accordance with the network resources a user requires, a switch can be configured to inform a network management station of any unauthorized access to network resources.

During inter VLAN communication, we can implement restrictions on a router to achieve it.

Flexibility and Scalability

By assigning switch ports or users to VLAN groups on a switch or group of switches, we gain flexibility to add only the users we want into that broadcast domain regardless of their physical location.

When a VLAN becomes to big, we can create more VLANs to keep broadcasts from consuming too much bandwidth.

Switching basics

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